Chen 2020 NPC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/14 samples (≥107 malignant cells; multiplicative count gap 107→36) intermix on one continuous manifold — mean iLISI 2.16 (≈50% of neighbourhoods cross the >2 threshold), a single connected component; the 39-cell detached speck was removed. No per-patient islands.

Gate 2 — module geography → GO (non-branching)

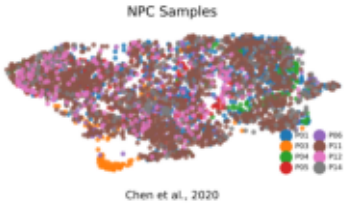
Gast & pSom do not resolve into two competing arms — both localise at the origin (high-decile mean pseudotime 0.22 each; Gast~pSom r=+0.00, no cross-repression). Crest is the inverse/terminal gradient (r=−0.78 vs Gast; high-decile pt 0.97; along pseudotime Gast ρ=−0.79, Crest ρ=+0.82). Non-branching.

Gate 3 — hub mutual exclusivity → NA

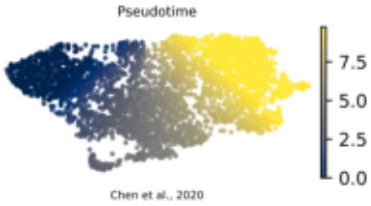
Single monotone Gast/pSom → Crest axis — no bifurcation and no Gast/pSom-overlap hub, so there is no branch pair to test for competitive cross-repression (branching-only gate; §2).

STEP FIGURES

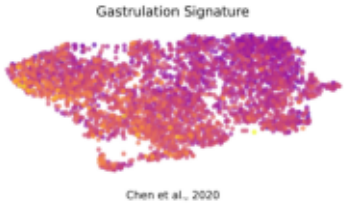
Gate 1: integration



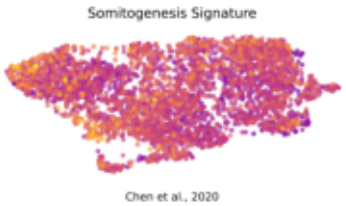
Trajectory / root



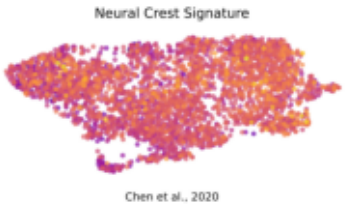
Gast



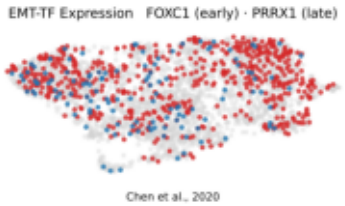
pSom



Crest



EMT-TFs



Dataset.

Nasopharyngeal carcinoma (Chen et al., 2020), 10x — 71,896 cells / 14 samples (3,763 malignant). Malignant only: 3,674 pass the ≥107-cell sample threshold; 3,635 (8 samples) enter the trajectory after the 39-cell detached speck is removed.

Verdict.

✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. The NPC malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. EMT-TFs FOXC1 (early/root) → PRRX1 (late/terminus); prototypical SNAI1/TWIST1 fall below the 3% detection floor and the panel is weakly organised (|ρ|<0.18).

Parameter choices

Min cells/sample	≥107 (keeps 8/14; gap 107→36)	**Integration**	Harmony(sample) θ=5; PCA 20PC
Island removal	radius-graph (−39 detached speck)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 0 = Gast/pSom-high, Crest-low	**Geometry**	non-branching curve (roundish, aspect 1.4)
Subtypes	markers + Chen 2020 (3CA mp_top)	**EMT-TFs**	FOXC1 (early) → PRRX1 (late)